MILITARY SPECIFICATION

CUTTING EDGES, SCRAPER

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fabricated and cast cutting edges for 8, 12, 18 and 24 cubic yard scrapers.

1.2 Classification. Cutting edges shall be of the following sizes us specified (see 6.2):

Size 1-8 Cubic Yard Scraper
Size 2 - 12 Cubic Yard Scraper
Size 3 - 18 Cubic Yard Scraper
Size 4- 24 Cubic Yard Scraper

Cutting-edges for the size 1, 2 and 4 shall be for the makes and models of scrapers as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of.

STANDARDS

Federal

FED. STD. NO. 151	- Metals, Test Methods.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.

FSC 3805

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MIL-STD-129	- Marking for Shipment and Storage.
MS39238	- cutting Edge, Scraper Bowl, Earth-moving, 18 Cubic Yard.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 0ther publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in ef'feet on date of invitation for bids or request for proposal shall apply.

SOCIETY OF AUTOMOTIVE ENGINEERS

SAE Handbook

(Application for copies should be addressed to the Society of Automotive Engineers, 485 Lexington Avenue, New York, N. Y. 10017.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., - Chicago, Illinois, 60606.)

AMERICAN TRUCKING ASSOCIATIONS, INC.

National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 P Street NW, Washington, D. C. 20036.)

3. REQUIREMENTS

3.1 Description. The cutting edges shall consist of three sections, two interchangeable end sections, and one reversible center section. The center section shall have two rows of holes to provide a drop center section.

3.2 Mechanical properties. The mechanical properties of the cutting edges shall conform to MS39238.

3.3 Dimensions and tolerances. Size 3 cutting edges shall conform to MS39238. Size 1, 2 and 4 cutting edges shall conform to SAE J737 for hole spacing and shall conform to SAE J738a for tolerances and cross section. The holes shall be for the correct size plow bolt in accordance with SAE J740 for countersunk square holes.

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3.4 Fabrication. Flame cutting to width or flame cut bevels shall not be permitted, Flame cutting to length shall be permitted provided the ends are annealed after cutting. Standard mill rolling tolerance shall apply on flatness, camber and bevel angle of cutting edges.

3.5 Workmanship. The rotting edges shall be free from sands injurious nonmetallic inclusions, segregation, cracks, seams, pipes, laps, fins, scale, sprues, pits, and other harmful extraneous material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection shall be classified as follows :

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, level I.

4.3.2 Examination.

4.3.2.3 Individual. Each cutting edge shall be examined for the critical defect specified in 4.4.1. Presence of a critical defect shall because for rejection.

4.3.2.2 Samples. Samples selected in accordance with 4.3.1 shall be . examined for the major defects specified in 4.4.1. AQL shall be 2.5 percent defective.

4.3.3 Tests. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 2.5 percent defective.

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4.4 Inspection procedure.

4.4.1 Examination. Cutting edges shall be examined for the following defects:

Critical.

1. Dimensional tolerances not as specified in 3.3.

Major.

101. Injurious nonmetal inclusions.
102. Segregation.
103. Cracks.
104* Seams l
105 l Pipes.
106. Holes not clean cut.
107. Lisps l
108. Fins.
109. Loose scale.

4.4.2 Tests.

4.4.2.1 Hardness test. The cutting edges (center and both end edges) shall be tested for Rockwell C hardness in accordance with FED. STD. No. 151, method 243.1. Take five hardness impressions across the width of the cutting edge on the side opposite the bevel. The core hardness shall be checked on at least one sample cutting edge of each lot of matting edges. Nonconformance to 3.2 shall constitute failure of this test.

4.4.2.E Charpy impact test. Cast cutting edges (center and both end edges) shall be tested for impact in accordance with FED. STD. No. 151, method 221.1. Impact shall be tested on at least one sample cutting edge of each lot of cutting edges. Nonconformance to 3.2 shall constitute failure of this test.

4.5 Inspection of-preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105 .

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent. defective.

110. Packing not as specified for level A.
111. Marking illegible, Incorrect, or incomplete.

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5. PREPARATION FOR DELIVERY

5.1 Preservation. Presentation shall be level A or C as specified (see 6.2).

5.1.1 Level A. Unpainted metal surfaces of the cutting edges shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.2 Level C. The cutting edges shall be preserved to prevent corrosion from the supplier to the initial destlnatton. The supplier's standard practice is acceptable when at fulfills these requirements.

5.2 Packing. Packing shall be level A or C as specified (see 6.2) e

5.2.1 Level A. The center edge section shall be laid flatwiththe two end edge sections placed end to end on top. Each of the end edge sections shall be secured to the center section with four ties of soft annealed wire having a minimum breaking strength of 300 pounds. For each end edge section, two ties of wire shall be placed through either at the holes nearest the center of the center edge section and through the end hole of the end edge section, and two ties of wire through the end hole of the center edge section and through the matching hole of the end edge section. The wires through the holes shall be placed in opposite directions and secured in a manner to prevent movement of the edges.

5.2.2 Level C. The cutting edges shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in compliance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules applicable to the mode of transportation.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The cutting edge blades are intended for use on scrapers for loading and leveling earth.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Make and model of scraper on which the size 1, 2, and 4 cutting edge will be mounted (see 1.2).
(d) Level of preservation and packing required (see 5.1 and 5.2).

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-C-52570 - Cutting Edges, Scraper

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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RESEARCH, DEVELOPMENT AND ENGINEERING DIRECTORATE
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